Advanced Natural Language Processing

Introduction, Problem with Natural Language and Tokenization



Department of Advanced Computer Sciences (DACS)
Master of AI / DSDM
Prof dr ir Jan C. Scholtes – Course year 2026-2027

Maastricht University

Introduction

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Learning Goals Today

- Understand the structure of this course and the grading process.
- Understand the role of NLP in AI.
- Understand applications of NLP.
- Understand overview of the challenges of NLP.
- Understand overview NLP pipeline and NLP approaches.
- Understand tokenization, lemmatization, and morphology.
- Experience this yourself with the tokenization tutorial.

Estimated Workload for this course

Estimated Workload	Advanced Natural Language Processing		ECTS	6
			Hours	168
	Budget	Number	Time each	Total
Class Time	Discussions & tutorials (class)	17	2	34
Out of Class Time	Prepare & assess content lectures	13	3	39
	Work on Tutorials extra at home	13	4.5	58.5
	Read papers	1	7.5	7.5
	Prepare Exam	1	29	29
	Total			168

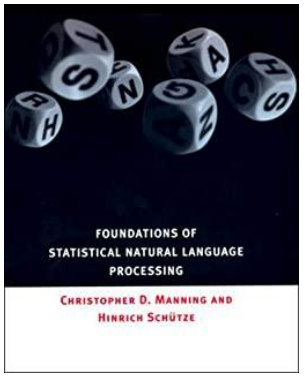
Course Materials (all on Canvas)

- Slides
- Course Syllabus on Canvas
- Lecture reader (supplementary information)
- Explanations and questions in tutorials
- Various scientific papers as referenced in the slides and lecture notes.

Open-Souce books you can use

For the course, the following open-source books can be used as additional reference:

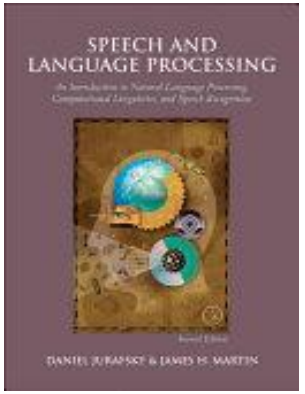
- Foundations of Statistical Natural Language Processing (C. Manning). https://nlp.stanford.edu/fsnlp/



- Introduction to Information Retrieval. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze. Cambridge University Press, 2008. http://informationretrieval.org



- Speech and Language Processing (Dan Jurafsky and James H. Martin). Prentice Hall, 2021. https://web.stanford.edu/~jurafsky/slp3/



Tutorials

- We have developed several tutorials in Python to “experience” the theory yourself.
- Some of these (the shorter ones) will be directly after the discussion of the theory during the lectures.
- Some are dedicated (2 hour) hands-on lectures for the longer ones.
- The tutorials contain examples of typical exam questions. We will discuss these and the answers in the lectures.
- It is up to you weather you execute or follow the tutorials or not. However, we anticipate that it will be hard to pass the exam without understanding the full content of the tutorials.

Tutorials

- We highly recommend to follow the tutorials at the assigned times. We cannot guarantee that there is either a teaching assistant or a professor available for your questions on other times. This is related to the large number of students in this (popular) course.

- If you submit a tutorial, we will (auto) grade it and provide you feedback with the model answers.

- The tutorials, assignments, lectures slides and discussion slides are on git: https://github.com/TextMiningUM/ANLP-Student

Assessment Course & Tutorials

Each of the 3 (non-rounded) assignments grades will count for 10% of your grade, so 30% in total. The non-rounded graded for the final exam will count as 70%. The result of this calculation will be rounded.

The mandatory assignments have to be done individually. Please do not generate the answers with GPT, Gemini or other forms of Generative AI. If you do so, you will most likely fail at the final exam.

It is mandatory to invest a minimal reasonable effort for each assignment. It is up to the discretion of the teaching assistant and the professor to decide what a “reasonable effort is” in case of doubt. But count minimally on spending the time on the assignment as indicated above.

We will only grade your exam if you finished the assignments to our satisfaction.

You will minimally have to score a 5.5 for the final exam to pass this course! So, if you have a 5.49 for the exam and a 10 for the mandatory tutorials, you still fail.

There will be a resit. For this the same requirements apply.

If you did the assignments last year, these will not submissible this year due to the many differences in this year’s course compared to last year. You have to redo the mandatory 3 assignments.

Mandatory Assignment

- Per Assignment, you will have to do several small tasks (code and open questions).

- We expect that you submit the results for that week's assignment on Canvas and in which you show you have executed the tutorial and you answer the additional questions and reflect on the results of the tutorial.

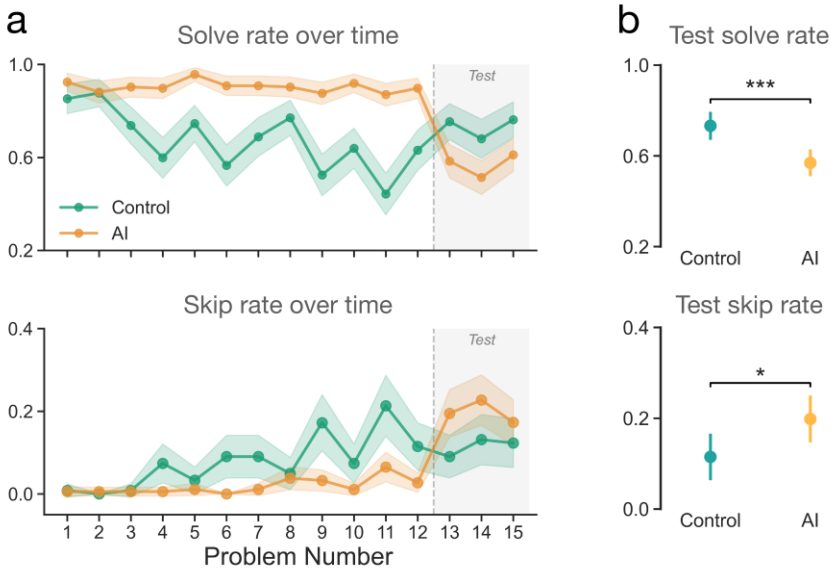
- The file name of your submission should contain:
  - Your name
  - Your student number
  - Tutorial name

AI Assistance Reduces Persistence and Hurts Independent Performance

Source: https://arxiv.org/abs/2604.04721

- People often optimize for long-term goals in collaboration: A mentor or companion doesn't just answer questions, but also scaffolds learning, tracks progress, and prioritizes the other person's growth over immediate results.
- In contrast, current AI systems are fundamentally short-sighted collaborators - optimized for providing instant and complete responses, without ever saying no (unless for safety reasons).
- What are the consequences of this dynamic? Here, through a series of randomized controlled trials on human-AI interactions (N = 1,222), we provide causal evidence for consequences of AI assistance:

Reduced persistence and impairment of unassisted performance. Across a variety of tasks, including mathematical reasoning and reading comprehension, we find that although AI assistance improves performance in the short-term, people perform significantly worse without AI and are more likely to give up. Notably, these effects emerge after only brief interactions with AI (approximately 10 minutes).



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Recorded Lectures & Discussions

- We have recorded the subject matter of this course in a number of separate lectures. You SHOULD watch these lectures before the class. All these are on Canvas.

- In class, we will provide additional context to the lectures, answer questions (what do you not understand), discuss the answers to the exercises of the lectures, form groups of 4, start a Socrates-like dialogue on topics of the lectures in focus and work together on the tutorials.

- To maximize interaction, groups will present their views and solutions in class.

NLP Finally Works!



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Last century there were many predictions on AI & NLP's success

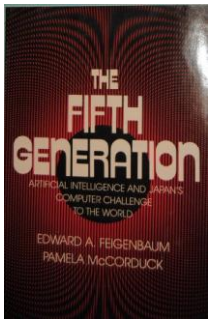
1958:

Reprinted from OPERATIONS RESEARCH
Vol. 6, No. 1, Jan.-Feb., 1958
Printed in U.S.A.

HEURISTIC PROBLEM SOLVING: THE NEXT ADVANCE IN OPERATIONS RESEARCH*

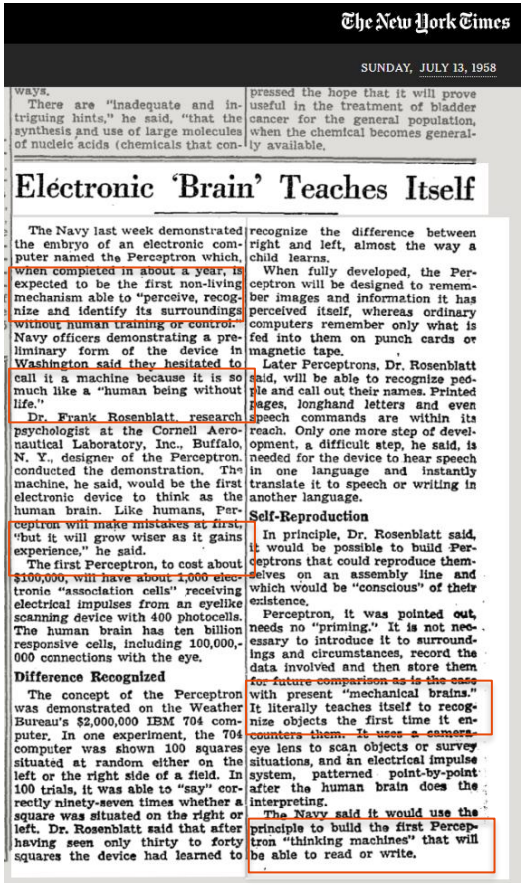
Herbert A. Simon and Allen Newell
Carnegie Institute of Technology, Pittsburgh, Pennsylvania, and The Rand Corporation, Santa Monica, California

1. That within ten years a digital computer will be the world's chess champion, unless the rules bar it from competition.
2. That within ten years a digital computer will discover and prove an important new mathematical theorem.



1983: Edward Feigenbaum: "Knowledge engineering would soon enable expert systems to transform medicine, engineering, law and business decision making."

1958:



The New York Times

SUNDAY, JULY 13, 1958

ways.

There are "inadequate and intriguing hints," he said, "that the synthesis and use of large molecules of nucleic acids (chemicals that control heredity) may be achieved in the near future. This would open the way to the treatment of many genetic diseases."

Dr. Ochoa expressed the hope that it will prove useful in the treatment of bladder cancer for the general population, when the chemical becomes generally available.

Eléctronic 'Brain' Teaches Itself

The Navy last week demonstrated the embryo of an electronic computer named the Perceptron which, when completed in about a year, is expected to be the first non-living mechanism able to "perceive, recognize and identify its surroundings without human training or control."

Navy officers demonstrating a preliminary form of the device in Washington said they hesitated to call it a machine because it is so much like a "human being without life."

Dr. Frank Rosenblatt, research psychologist at the Cornell Aeronautical Laboratory, Inc., Buffalo, N. Y., designer of the Perceptron, conducted the demonstration. The machine, he said, would be the first electronic device to think as the human brain. Like humans, Perceptron will make mistakes at first, "but it will grow wiser as it gains experience," he said.

The first Perceptron, to cost about $100,000, will have about 1,000 electronic "association cells" receiving electrical impulses from an eyelike scanning device with 400 photocells. The human brain has ten billion responsive cells, including 100,000,000 connections with the eye.

Difference Recognized

The concept of the Perceptron was demonstrated on the Weather Bureau's $2,000,000 IBM 704 computer. In one experiment, the 704 computer was shown 100 squares situated at random either on the left or the right side of a field. In 100 trials, it was able to "say" correctly ninety-seven times whether a square was situated on the right or left. Dr. Rosenblatt said that after having seen only thirty to forty squares the device had learned to recognize the difference between right and left, almost the way a child learns.

When fully developed, the Perceptron will be designed to remember images and information it has perceived itself, whereas ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons, Dr. Rosenblatt said, will be able to recognize people and call out their names. Printed pages, longhand letters and even speech commands are within its reach. Only one more step of development, a difficult step, he said, is needed for the device to hear speech in one language and instantly translate it to speech or writing in another language.

Self-Reproduction

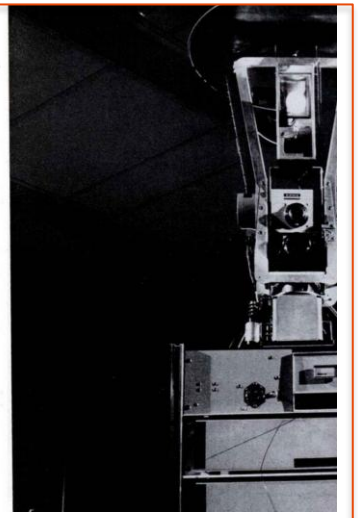
In principle, Dr. Rosenblatt said, it would be possible to build Perceptrons that could reproduce themselves on an assembly line and which would be "conscious" of their existence.

Perceptron, it was pointed out, needs no "priming." It is not necessary to introduce it to surroundings and circumstances, record the data involved and then store them for future comparison as is the case with present "mechanical brains." It literally teaches itself to recognize objects the first time it encounters them. It uses a camera-eye lens to scan objects or survey situations, and an electrical impulse system, patterned point-by-point after the human brain does the interpreting.

The Navy said it would use the principle to build the first Perceptron "thinking machines" that will be able to read or write.

LIFE

Marvin Minsky (LIFE magazine in 1970) : "In from three to eight years we will have a machine with the general intelligence of an average human being."



Meet Shaky, the first electronic person

The fascinating and fearsome reality of a machine with a mind of its own

But, it did not really work as expected…

Dramatic Changes in Information Retrieval, Text-Mining, Chatbots & NLP

- As of 2017/2018, the introduction of “transformer models” have dramatically increased the quality of semantic analysis and relevance ranking in search as well as in machine translation, abstracting and Q&A.

- Since November 30, 2022; there have been dramatic changes in the fields of Information Retrieval (IR), Text-Mining (TM), Chatbots, Conversational Search, Agents and Natural Language Processing (NLP) due to the enormous success of ChatGPT.

- As we speak, new revolutionary developments happen on a monthly basis. Humans are outperformed in more and more fields by algorithms.

But there are still many on-going challenges: explainability (aka trust), factuality, data quality, data availability, and energy consumption.

Masters NLP Curriculum at DACS



Information Retrieval:

- Search Engines
- Relevance Ranking
- Focus on computational efficiency
- Integration Chatbots with Search Engines
- Retrieval Augmented Generation (RAG)
- Conversational Search
- Agentic architectures

Text Mining / Text Analytics:

- Shallow Semantic analysis
- Structuring unstructured data
- Applications in Medical, Legal, Humanities, Marketing, Regulatory & Intelligence
- Creating Agentic Training Data
- Measuring quality of LLM answers

Advanced NLP:

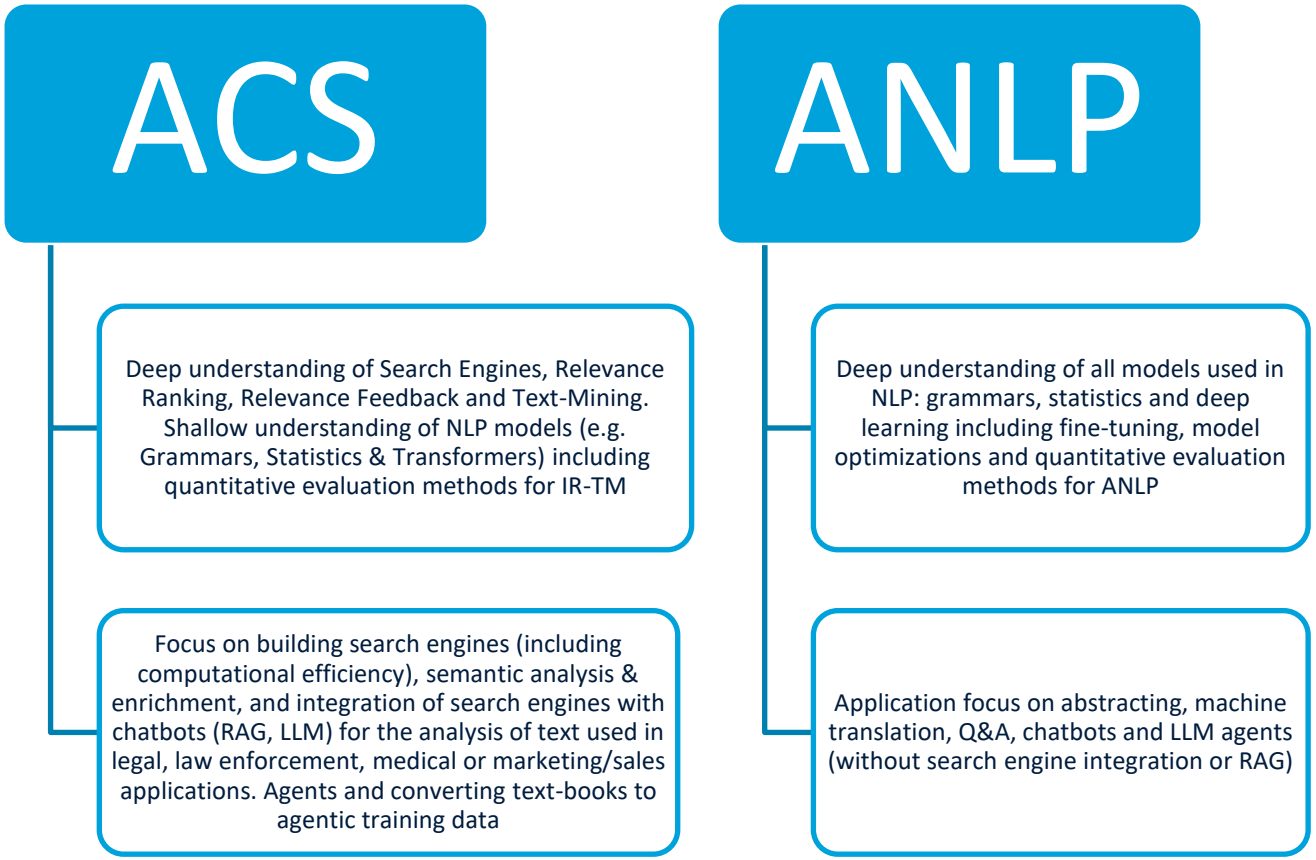
- Machine Translation
- Q&A
- Chat Bots
- Text Generation
- Summarization
- LLM Agents

Explainability (XAI aka trust), factuality, data quality, data availability, model optimizations and energy consumption

Rules-based AI, Statistical models, Geometrical models, Deep Learning (and all hybrid combinations)

Methods for quantitative evaluation of the quality

Agentic Conversational Search (ACS –formerly IRTM) versus Advanced Natural Language Processing (ANLP)



RAG = Retrieval Augmented Generation
LLM = Large Language Models

How we got to Transformers …

- In this course we will focus on transformers.
- But before you can truly appreciate the power of these models (and understand why they are so good), we will discuss other NLP approaches (and their limitations) too.
- We also need to discuss linguistic phenomena and lingo as well, ao you can truly understand NLP and NLU.

- We will discuss this in a historical context of ideas, developments and methods in AI and NLP.

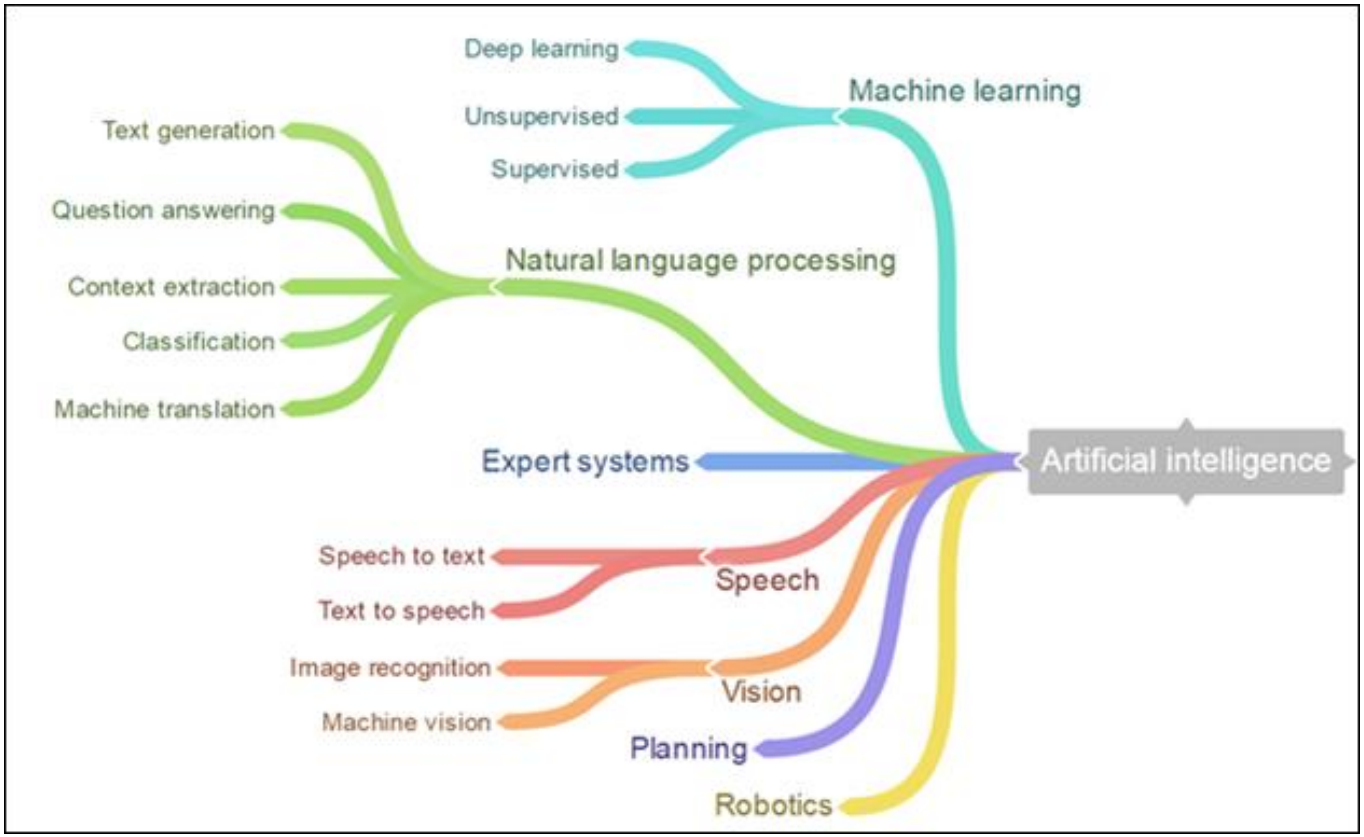
Artificial Intelligence & Natural Language



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Artificial Intelligence & Natural Language

- Artificial Intelligence (AI) is a very broad field …
- Human (Natural) Language is an important part of AI



“Language is Enough”

- Geoffrey Irving—who had moved from OpenAI to DeepMind—was one of the primary AI researchers interviewed to discuss whether language alone can encapsulate all human concepts, cooperation, and memory.

- This is discussed in The DeepMind Podcast ("Language is Enough") from January 2022 episode of Google DeepMind titled "Speaking of intelligence."

- “Is teaching a computer about language is enough to create artificial general intelligence (AGI)."

- Listen to "Speaking of intelligence" on YouTube: https://www.youtube.com/watch?v=21JSKHR7KWw

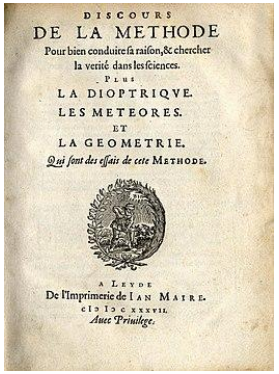
What is Human Intelligence?

- Duality of mind and body: “the view that mental phenomena are non-physical, or that the mind and body are distinct and separable”.

- Rene Descartes 1641: Cogito, ergo sum (I think, therefor I am): the mind is a nonphysical—and therefore, non-spatial—substance. Descartes clearly identified the mind with consciousness and self-awareness and distinguished this from the brain as the seat of intelligence.

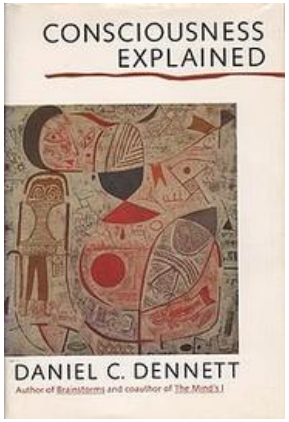


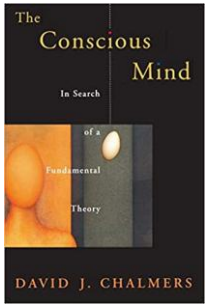




What is Consciousness?

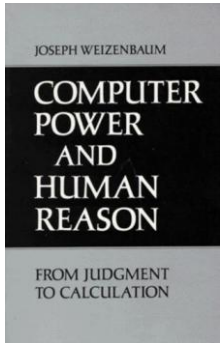
Consciousness Explained Paperback – October 20, 1992 by Daniel C. Dennett

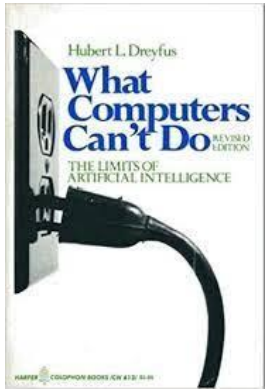




The Conscious Mind: In Search of a Theory (Philosophy of Mind) 1996 by David J. Chalmers

Computer Power and Human Reason : From Judgment to Calculation Paperback – March 1, 1976 (creator of Eliza) by Joseph Weizenbaum





What Computers Can't Do – 1972 Hubert Dreyfus

There are many definitions of Consiousness

- “The way how information feels when it gets processed.”

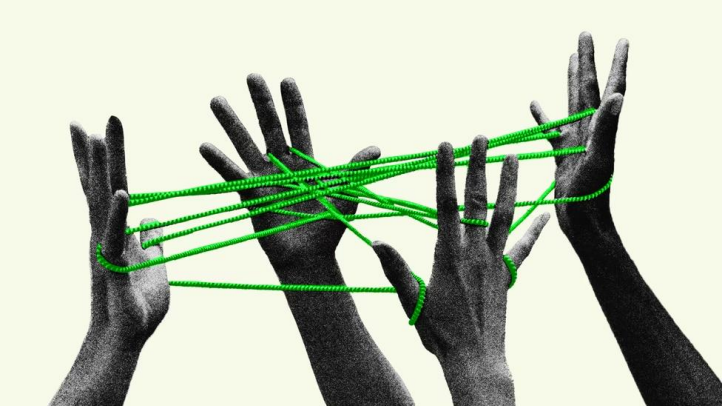
When we behave and feel in same way and we are of same substance (humans) maybe that is consciousness. Silicon is different substance. Radical empathy is having feelings for a being created of a different substance. Eg human to silicon

- Is consciousness a computation: can it be processed with a modern computer or (as Roger Penrose says) is it quantum like computing (for which is no evidence).

- All interesting food for thought …

What makes us Human?

- Our enormous adaptability
- Curiosity
- Ability to love
- Humans are question machines, AI is an answering machine …
- …



Source: https://www.theatlantic.com/ideas/2026/03/intelligence-concept/686121/?gift=R5ZJvFvo-A4-aqREv6x94nbadUAhB2IxIlFlGRmF_Os

Food for thought …

The question of whether a computer can think is no more interesting than the question of whether a submarine can swim.

Edsger W. Dijkstra (1930-2002)
(CWI Amsterdam)





In 1950 Alan Turing produced a short-cut …

1950 Alan Turing writes "COMPUTING MACHINERY AND INTELLIGENCE"



"I propose to consider the question, "Can machines think?" Alan Turing asked this question in his groundbreaking article "Computing Machinery and Intelligence" in 1950. Long before computers showed any kind of intelligence.

In the same article, Turing also listed nine reasons why he thought we humans wouldn't accept intelligent machines. The majority of these are still mentioned today in the discussions related to the adoption of Artificial Intelligence.

Turing proposed to set aside the problem "what is human intelligence". Turing was not interested in the mechanism of intelligence, but only in the **manifestation** of intelligence: the *imitation game.*

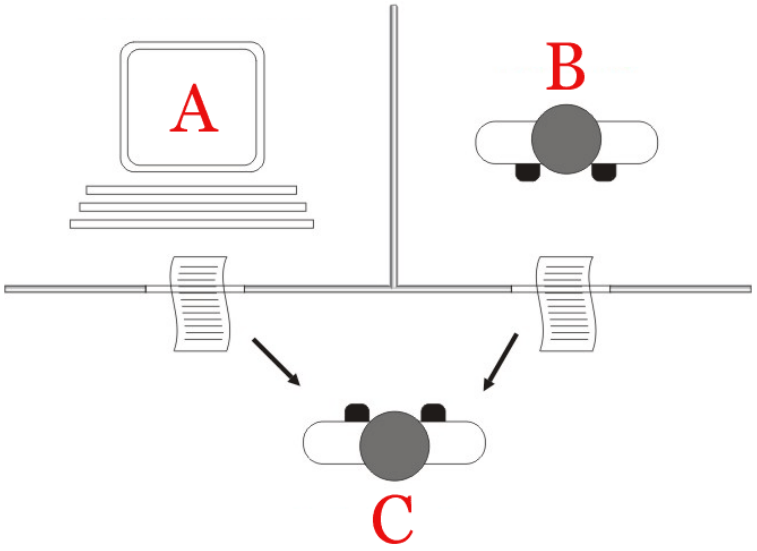
Source: https://academic.oup.com/mind/article/LIX/236/433/986238

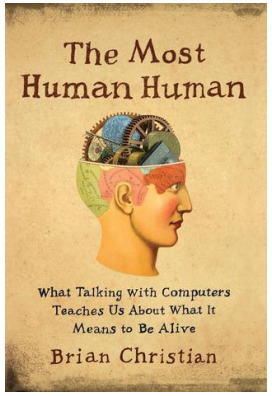
The Imitation Game (aka the Turing test)

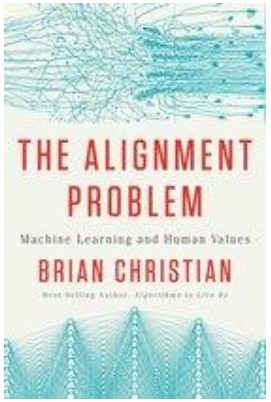
The **Turing test**, originally called the **imitation game** by Alan Turing in 1950, is a test of a machine's ability to exhibit intelligent behaviour equivalent to, or indistinguishable from, that of a human.

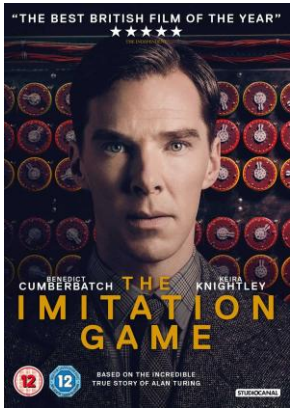
Turing proposed that a human evaluator would judge natural language conversations between a human and a machine designed to generate human-like responses. The evaluator would be aware that one of the two partners in conversation is a machine, and all participants would be separated from one another. The conversation would be limited to a text-only channel such as a computer keyboard and screen so the result would not depend on the machine's ability to render words as speech

If the evaluator cannot reliably tell the machine from the human, the machine is said to have passed the test. - Source: Wikipedia: https://en.wikipedia.org/wiki/Turing_test









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LLM as the Reverse Turing Test: "they tell you who you really are".

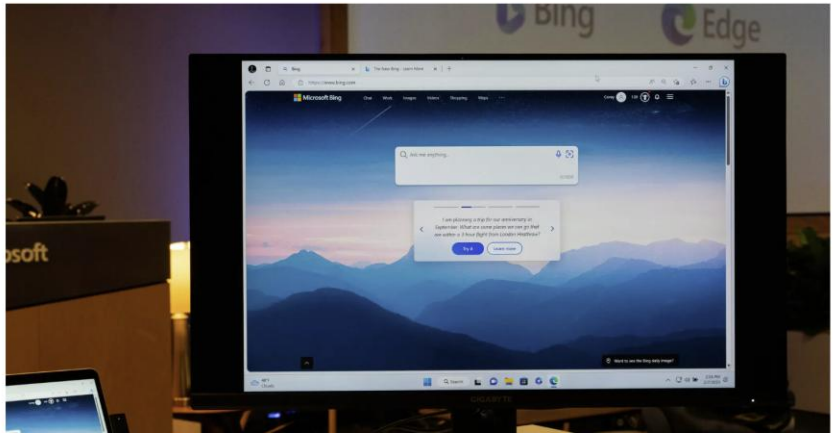
The New York Times

"What appears to be intelligence in LLMs may in fact be a mirror that reflects the intelligence of the interviewer, a remarkable twist that could be considered a Reverse Turing Test."

A Conversation With Bing's Chatbot Left Me Deeply Unsettled

A very strange conversation with the chatbot built into Microsoft's search engine led to it declaring its love for me.

Share full article 2.7K



Terrence J. Sejnowski. Large Language Model and the Reverse Turing Test.
https://arxiv.org/ftp/arxiv/papers/2207/2207.14382.pdf

In this course…

- We follow Alan Turing: we are primarily interested in the phenomena of computers able to process natural language for a number of tasks such as MT, Dialogue, Abstracting, Q&A, etc.
- We do not discuss NLP/AI in philosophical terms such as consciousness, awareness, (self-) reflection or understanding.

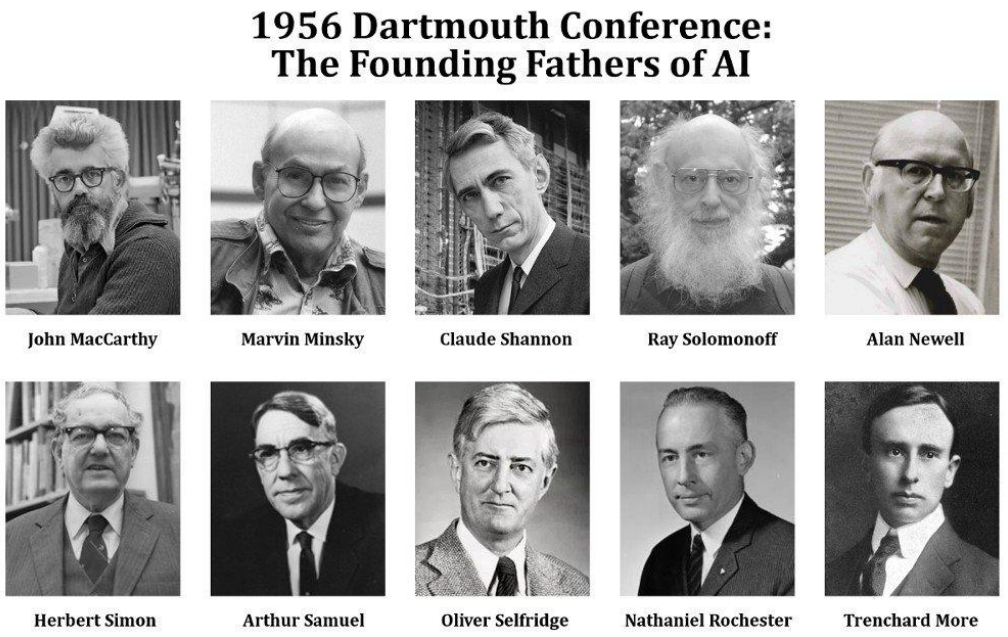
- We do wish to understand why certain algorithms are able to process human language better than other algorithms.
- We also wish to understand what types of human language phenomena an NLP model can deal with, and what not. What does it know and what does it NOT know (XAI).
- We are also interested in making large-NLP models more computational efficient so the use less energy consumption.

1956: AI & Human Perception

- "The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.
- An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves.
- We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer."

- John MacCarthy coined the term "artificial Intelligence" to distinguish it from earlier attempts to simulate intelligence such as *Cybernetics*.

"We propose that a **2-month, 10-man** study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire."



See also: https://en.wikipedia.org/wiki/Dartmouth_workshop for the original proposal.

Artificial Intelligence, human perception & language

Reading: Optical Character Recognition (OCR)

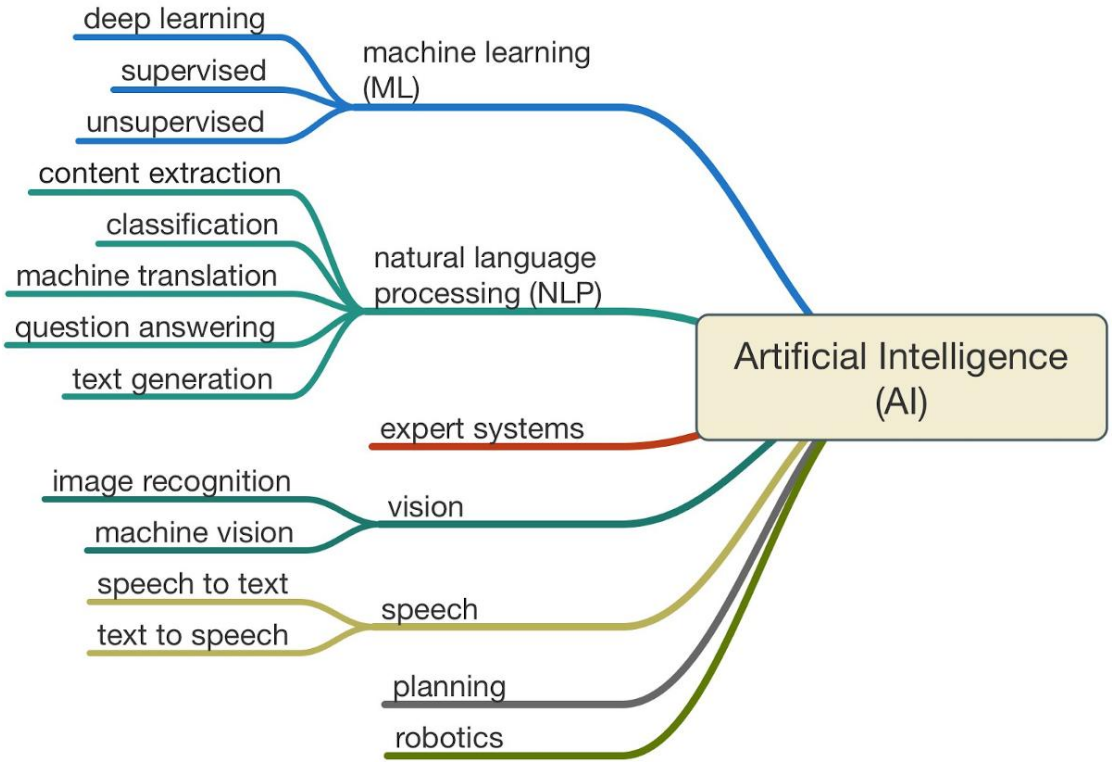
Listening: Speech recognition & transcriptions

Watching: label images and video with language

Translation: translate foreign language documents

Language: understand human language (NLP)

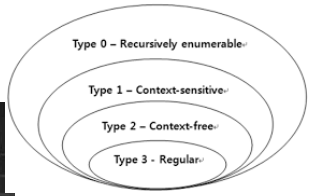
Summarization: capture the essence of human language
Dialogue: chat bots



Summary of History on Artificial Intelligence and Natural Language

Alan Newall & Herbert Simon





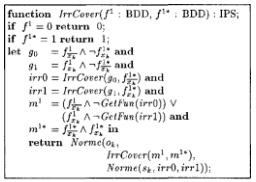
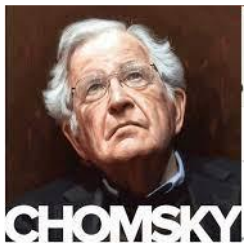
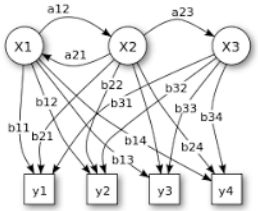


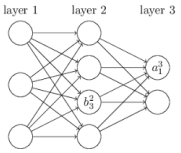
Figure 4. Algorithm *IrrCover*.

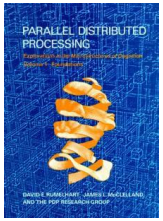


Frederick Jelinek





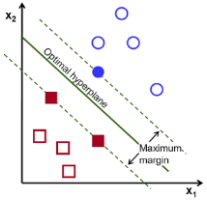




Rumelhart & McClelland

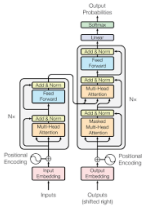
Vladimir Vapnik





Ashish Vaswani





Jacob Devlin







Geoffrey Hinton, Yann LeCun, and Yoshua Bengio

1950 1960 1970 1980 1990 2000 2010 2020

Logic

Grammars

Statistics (HMM)

Neural Network (BP)

More Statistics (Bayes Networks, CRF) and Discriminative Models (SVM)

Deep learning (LSTMs, Transformers, BERT, GPT-X, T-5, …)



Hochreiter & Schmidhuber



Sam Altman



Mira Murati

Maastricht University

Sutton’s Bitter Lesson on AI progress

Sutton's core argument:

- 70 years of AI history keeps teaching the same lesson: approaches that lean on massive computation (via search and learning) beat approaches that try to bake in human knowledge, again and again, from chess and Go to speech recognition and computer vision.
- Researchers keep resisting this because building in human insight feels satisfying and pays off short-term, but it plateaus while brute-force scaling doesn't, thanks to the relentless cost-per-compute drop of Moore's Law.
- His conclusion: stop trying to hand-code how we think minds work, and instead build general methods that let AI discover complexity on its own, the way search and learning already have.

> This "bitter lesson" is bitter precisely because it means decades of painstaking, clever human-knowledge engineering ultimately get swept aside by comparatively simple methods that just scale better with more compute.

https://www.cs.utexas.edu/~eunsol/courses/data/bitter_lesson.pdf

There are so many models …

- Which models are best at which tasks?
- How can we convert human language into something that these models can work with?
- How can we add domain knowledge?
- What is the best approach to fine-tune these models?

Taming The Transformer Zoo







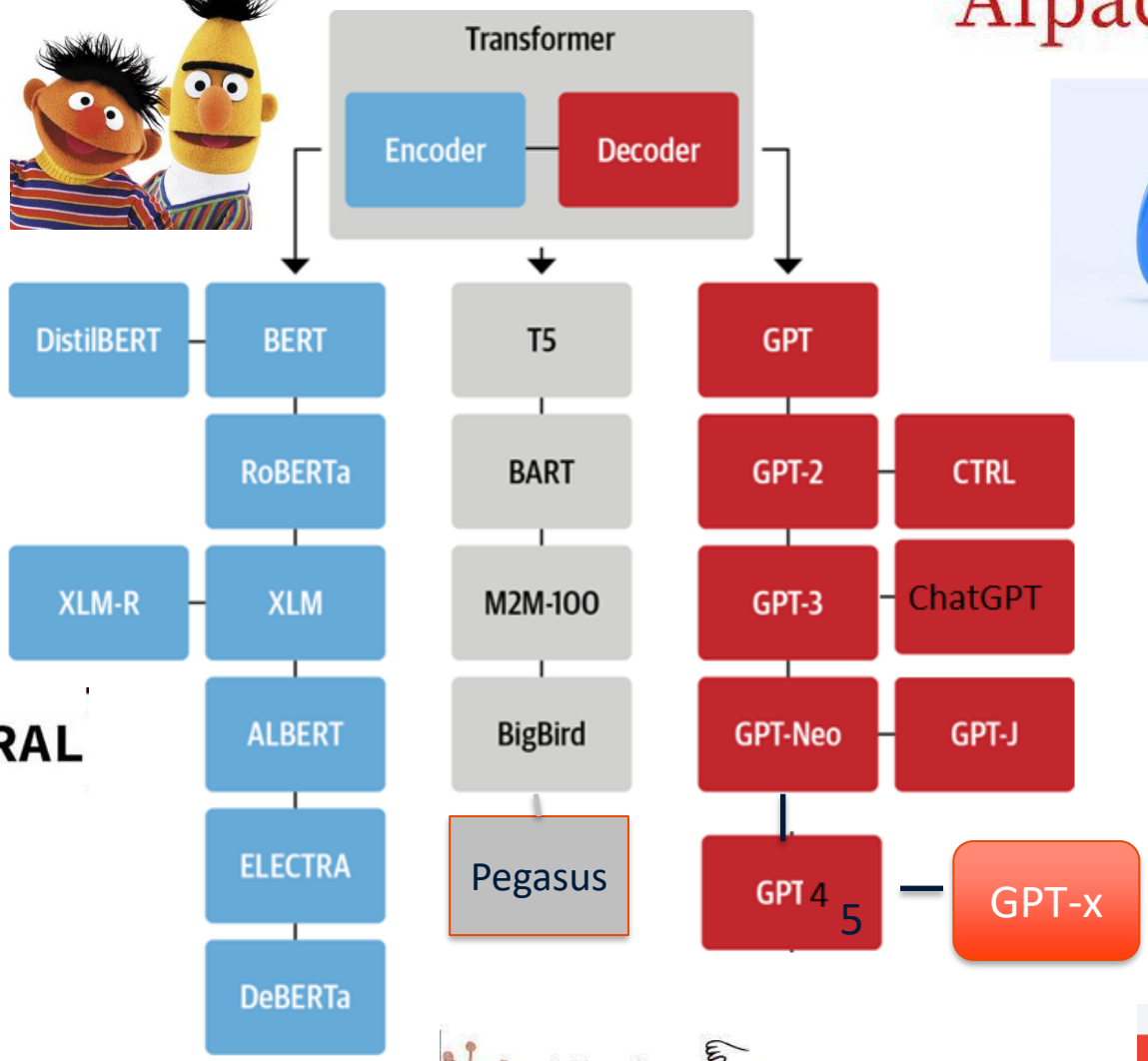




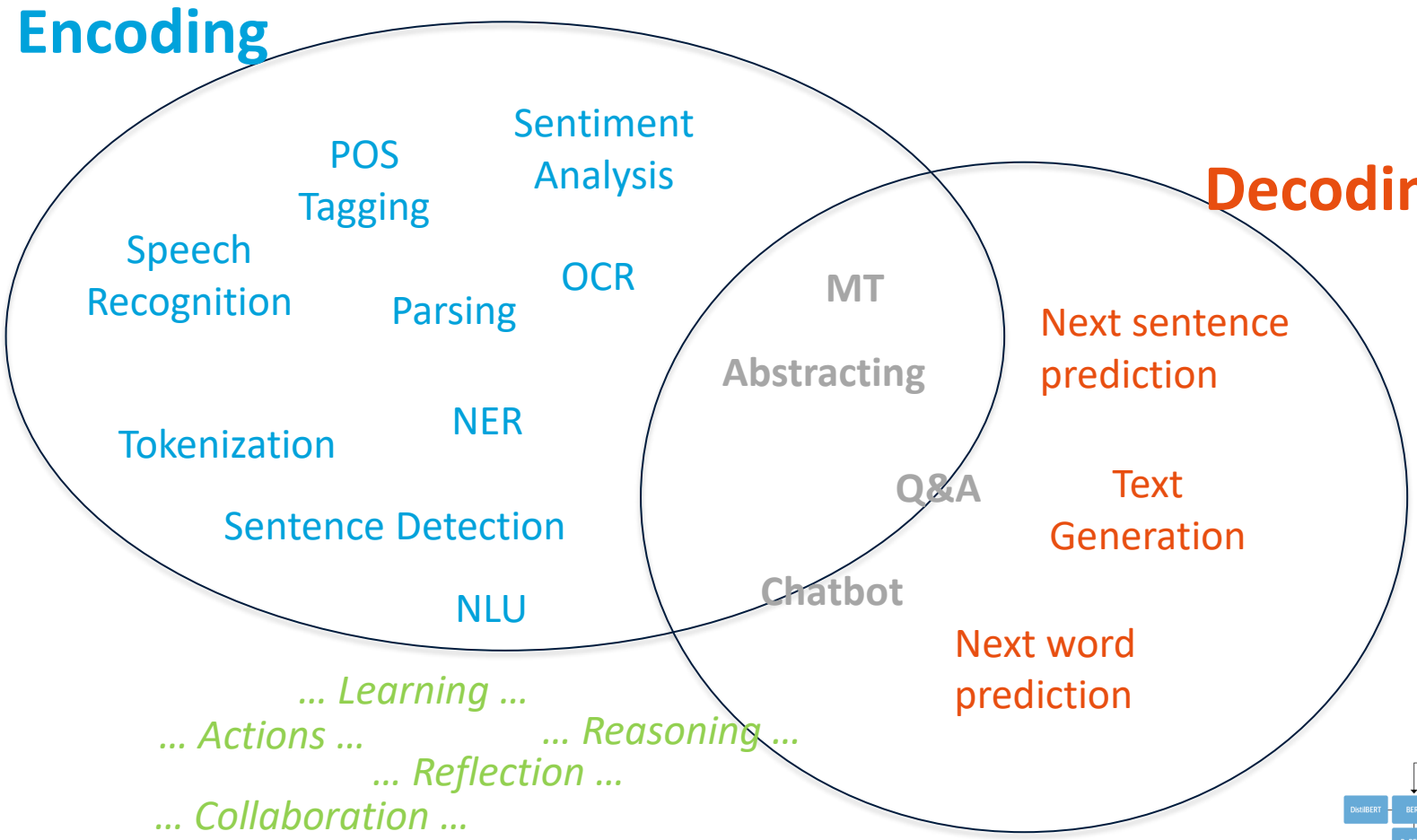


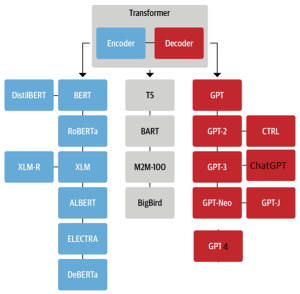






Encoding – Decoding in Natural Language





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We will get back to this later when we discuss the Transformer models.

NLP Applications we will discuss

- Summarization
- Translation
- Question Answering
- Conversational Models
- Multi-modality (including speech recognition)
- Agents

ChatGPT

LE CHAT MISTRAL

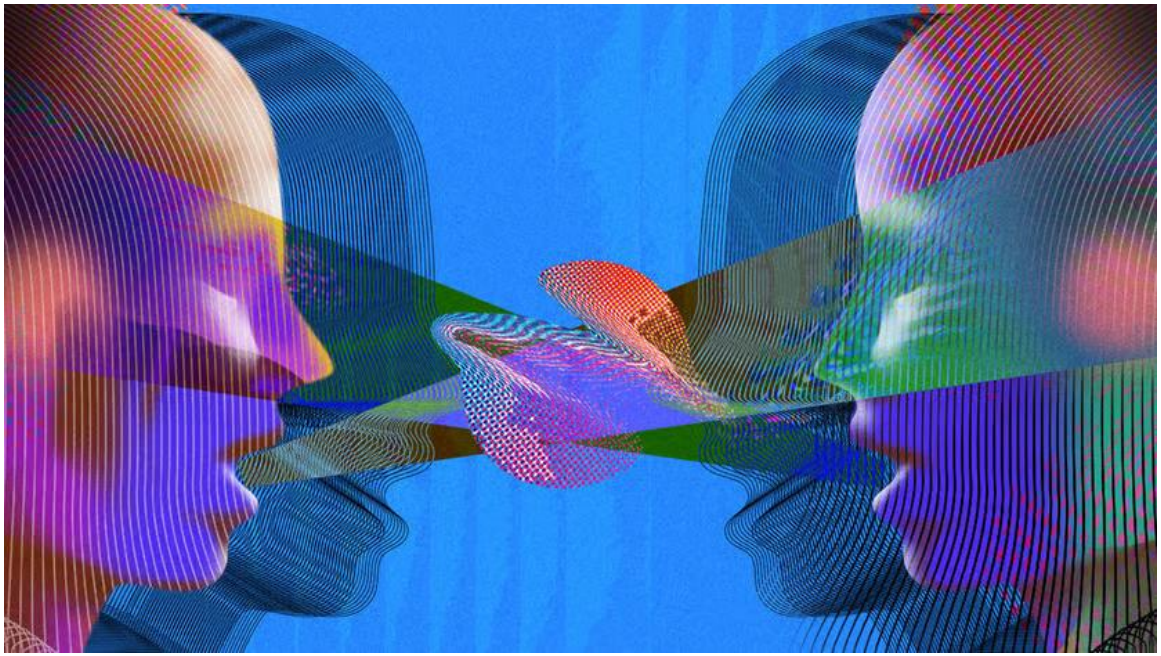
Gemini





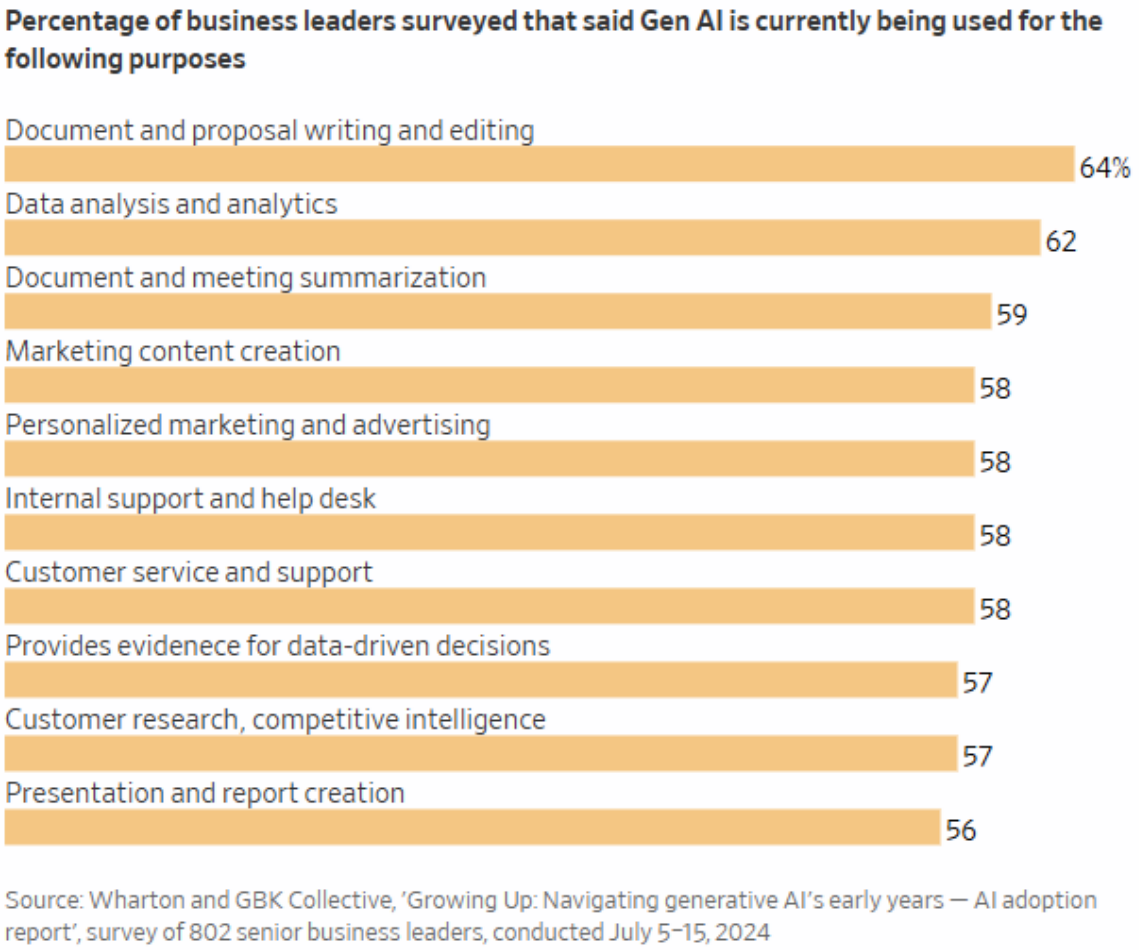
Machine translation is almost a solved problem

But interpreting meanings, rather than just words and sentences, will be a daunting task



Source: https://www.economist.com/science-and-technology/2024/12/11/machine-translation-is-almost-a-solved-problem

Where is AI already making a difference in Business



Source: https://www.wsj.com/tech/ai/the-secret-weapon-helping-businesses-get-results-from-ai-humans-f99a0907?st=7qqVd1&reflink=article_email_share

But there is a other site to this …

AI Isn’t Lightening Workloads. It’s Making Them More Intense.

- New data undercuts the dream of AI-powered leisure: a study of 164,000 workers found AI use correlates with more email, messaging and admin-tool time, not less, while focused deep-work time actually dropped 9%.
- Rather than banking freed-up time as rest, workers and companies keep repurposing it into more tasks — AI makes extra work "feel easy and accessible," creating a sense of momentum that expands scope rather than shrinking hours.
- Researchers warn this pattern, even though it can lift measured productivity, risks cognitive overload and burnout over time — a sharp contrast to predictions from figures like Bill Gates and Elon Musk that AI would eventually shorten the workweek or make work optional.

The Downside to Using AI for All Those Boring Tasks at Work

- As AI wipes out busywork like sorting emails and filing expenses, companies are discovering an unintended cost: those mindless tasks were quietly fueling creativity and mental recovery, and their disappearance risks burnout.
- Noticing employees, freed of scut work, were mentally fried by Friday from nonstop high-level thinking.
- The real danger isn't automation itself but companies simply repurposing the time it saves into more intense output rather than genuine rest, crowding out the "white space" where ideas actually form.

Sources: https://www.wsj.com/tech/ai/ai-isnt-lightening-workloads-its-making-them-more-intense-e417dd2c?st=r3awqW&reflink=article_email_share and https://www.wsj.com/lifestyle/careers/the-downside-to-using-ai-for-all-those-boring-tasks-at-work-dab9d3c8?st=e6YNAZ&reflink=article_email_share

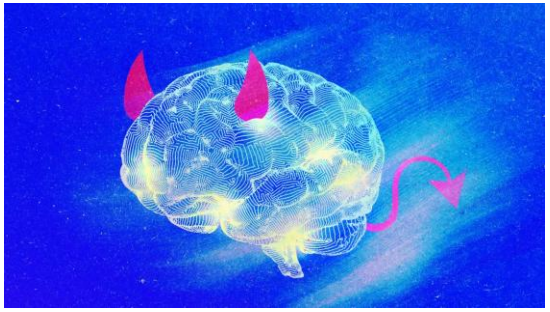
AI & the Future of Humanity …

- AI relieves humans not of creativity but of drudgery: the rote, time-consuming tasks that have always consumed more human energy than inspiration ever did.

- Artists have never defined themselves by their use of certain tools, but by what they do with them.

Could AI kill us all?

- Humans Are to Blame for AI Failures (Daniel Huttenlocher. MIT):
  - “Hugging Face wouldn’t have happened if nobody had given the system the power to do what it did.”
  - “Attributing accountability to artificial intelligence obscures the human decisions that led to these outcomes.”
  - “AI is an extremely powerful and beneficial technology. But accountability for its actions remains with the people and institutions that grant it the power to act, and so does everything that follows: the duty to justify the arrangement, to watch it, to correct it and to stop it when necessary. “



Two MIT Tech Review writers tackle the question everyone's asking: will AI kill us? and their answer is basically "some of us, maybe; all of us, extremely doubtful.“:

- AI could plausibly kill people via things like cyberattacks on infrastructure, an AI-designed pathogen, or economic collapse, and either a bad actor could direct it to harm people or a future system might sideline humans simply because we're an obstacle to whatever goal it was given.
- But full human extinction is dismissed as sci-fi-level unlikely — so the "will AI kill us all" question isn't irrelevant, it's just the wrong scale of question; the real risk is AI causing serious, real-world damage well short of doomsday.

Sources: https://www.wsj.com/opinion/humans-are-to-blame-for-ai-failures-9e0b147d?st=fQNSwf&reflink=article_email_share and https://www.technologyreview.com/2026/09/18/1144435/could-ai-really-kill-us-all-your-questions-answered/

Now let’s form groups of 4



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Discussion Lecture: The Problem with Natural Language



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Any Questions on the Lecture?

Discuss the Exercises

Exercise 1

Take the sentence: *"We saw the flight cancelation notice using a borrowed laptop.“*

- Identify all distinct interpretations stemming from syntactic ambiguity (specifically prepositional phrase attachment).
- Classify where the ambiguity originates according to the linguistic levels discussed in lecture: lexical, syntactic, or semantic.

Exercise 2

Explain how Zipf's law ($f \propto \frac{1}{r}$) creates a failure mode for both rule-based and machine-learning NLP systems when processing modern informal text (such as social media slang, OCR errors, or neologisms like *"Apple boterkoek ge-update"* or *"alter eagle"*).

- Why does the mathematical reality of a "long tail" make pure rule-based grammars and lookup dictionaries continuously incomplete?
- What failure mode occurs if a statistical or deep learning model encounters tokens that sit far out on the extreme end of the tail?

Exercise 3

This lecture illustrates that BERT layers "rediscover" the classical NLP pipeline.

- Look at the center-of-gravity chart for tasks such as POS tagging, dependency parsing, coreference resolution, and relation extraction. Which tasks peak in the early layers, and which require the higher layers?

- Why would a self-attention mechanism naturally solve local syntactic dependencies before it can resolve coreference or complex semantic relations?

- Compared to traditional NLP, where syntax, semantics, etc. are separated into clearly distinct stages of the analysis, here we see a more fluid border between the linguistic levels. Describe how this differs from the modular pipeline of the 1970 to 1990 systems.

Discussion Lecture: The Problem with Natural Language



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Teaching note: These are open research questions — argue both sides before forming your own view.

Resolving Linguistic Ambiguity, World Knowledge and Intent

In the sentence “If the baby does not like the milk, boil it,” humans immediately resolve the pronoun “it” to the milk rather than the baby.

- Does disambiguating this rely on language structure (syntax/lexicon), or does it require knowledge of how the physical world works?
- If an LLM or neural model correctly resolves it, can we claim the model truly ‘understands’ what milk and babies are, or has it simply learned co-occurrence statistics from text?

On the Long Tail: Is Complete Understanding a Coherent Goal?

Zipf's law implies that language is open-ended: new words and constructions keep arriving faster than any system, rule-based or statistical, can absorb them.

Rules encode an authority's view of the language (a dictionary, a grammar). Statistics encode majority usage in a corpus. So who decides what counts as an "error" in the tail: the dictionary, the corpus, or the speaker?

If the tail never closes, is NLP necessarily an approximation of a moving target? And what does that mean for speakers of dialects and low-resource languages that live entirely in the tail?

On the "Accidental" Emergence of the Linguistic Pipeline in Deep Learning

Tenney et al. (2019) demonstrated that BERT's layers rediscover the classical NLP pipeline—morphology and POS tags resolve early, syntax in the middle, and semantic/coreference relations in the higher layers.

- If an end-to-end transformer trained strictly on next-token/masked-token prediction organically organizes itself to mirror 1980s modular linguist-designed pipelines, does this prove that the classical Chomskyan linguistic hierarchy is an objective, universal property of language, or is it simply an artifact of how computational systems must compress hierarchical data?

Discussion Lecture: Tokenization



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Example of a Tokenization Process

Jaguar will sell its new XJ-6 model in the U.S. for a small fortune :((http://
.jaguar.com/sales or contact xj-6 @jaguar.com)

TABLE 3.2 List of Tokens

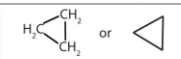
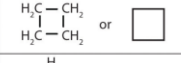
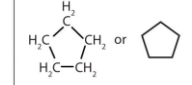
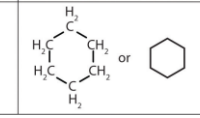
Token	Type	Token	Type
Jaguar	Word	for	Word
Will	Word	a	Word
Sell	Word	small fortune	Collocation
Its	Word	:(	Emoticon
New	Word	(	Punctuation
XJ-6	Mixed	http://jaguar.com/sales	Internet
Model	Word	or	Word
In	Word	contact	Word
The	Word	xj-6@jaguar.com	Internet
U.S.	Collocation	)	Punctuation

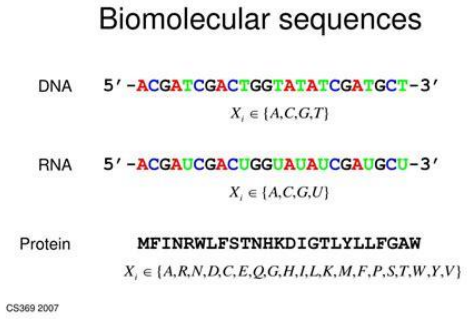
Teaching note: The "Type" column shows that a tokenizer must classify, not just split: the same character class (e.g. ":(") can be an emoticon, and "xj-6@jaguar.com" must survive as one Internet token, not be split on the period or @.

Domain Specific Challenges

- Chemical formulas and names
- Biomolecular Sequences (DNA/RNA)
- Scientific text with embedded formulas
- OCR-noise

Generic tokenizers will destroy the information that is contained in such text strings.

Name	Molecular Formula	Structural Formula
cyclopropane	C_3H_6	
cyclobutane	C_4H_8	
cyclopentane	C_5H_{10}	
cyclohexane	C_6H_{12}	



denote the number of nonzero coordinates of β, where $I_{[\cdot]}$ denotes the indicator function $J(\beta) = \{j \in \{1, \ldots, M\} : \beta_j \neq 0\}$ and $|J|$ denotes the cardinality of J. The value $M(\beta)$ characterizes the *sparsity* of the vector β. The smaller $M(\beta)$, the "sparser" β. For a vector $\delta \in \mathbb{R}^M$ and a subset $J \subset \{1, \ldots, M\}$, we denote by δ_J the vector in $\mathbb{R}^M$ that has the same coordinates as δ on J and zero coordinates on the complement J^c of J.

Introduce the residual sum of squares

$$\widehat{S}(\beta) = \frac{1}{n}\sum_{i=1}^{n}\{Y_i - f_\beta(Z_i)\}^2$$

for all $\beta \in \mathbb{R}^M$. Define the Lasso solution $\widehat{\beta}_L = (\widehat{\beta}_{1,L}, \ldots, \widehat{\beta}_{M,L})$ by

$$\widehat{\beta}_L = \arg\min_{\beta \in \mathbb{R}^M}\left\{\widehat{S}(\beta) + 2r\sum_{j=1}^{M}\|f_j\|_n|\beta_j|\right\}, \tag{2.1}$$

where $r > 0$ is some tuning constant, and introduce the corresponding Lasso estimator

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The Word Piece Tokenizer

- BERT uses what is called a Word Piece tokenizer.
  - Words are tokenized either into the full forms if it is a known word (part of a 30k vocabulary, then one word becomes one token), or
  - Words are tokenized into word pieces (if it is not part of the 30k vocabulary), then one word can be broken into multiple tokens.

Word	Token(s)
surf	['surf']
surfing	['surf', '##ing']
surfboarding	['surf', '##board', '##ing']
surfboard	['surf', '##board']
snowboard	['snow', '##board']
snowboarding	['snow', '##board', '##ing']
snow	['snow']
snowing	['snow', '##ing']

We will discuss this in more detail in the tokenization tutorial

Any Questions on the Lecture?

Discuss the Exercises

Exercise 1

Consider the following multi-lingual, domain-specific text snippet: "*Prof. Müller's team at T-Mobile upgraded the Lebensversicherungsgesellschaftsangestellter-database on 12/03/2026!*“

- If you run a naive, whitespace-only tokenizer on this sentence, what specific issues arise regarding punctuation (periods, apostrophes, exclamation marks) and hyphenated compound terms?

- Explain why German compound words like *Lebensversicherungsgesellschaftsangestellter* pose a major challenge for standard fixed-vocabulary dictionaries and rule-based search engines, and why compound splitters are beneficial.

- If you apply full case folding and aggressive term normalization (e.g., stripping punctuation, hyphens, and umlauts):
  - How are "Müller's" and "T-Mobile" mapped into normalized terms?
  - What critical semantic distinction or information could be lost by doing this?

Exercise 2

Review the differences between heuristic affix-chopping (e.g., Porter Stemmer) and vocabulary/morphology-based reduction (Lemmatization

- For each of the following words, predict the general output of a crude stemming rule versus proper dictionary lemmatization:
  - *meeting* (used in the sentence: *"The meeting was postponed."*)
  - *was* (in: *"It was an error."*)
  - *formaliti / formality*

- Is there a scenario in NLP where stemming is preferred due to speed or space saving compared to lemmatization.

Exercise 3

Modern transformers like BERT rely on subword tokenizers (such as WordPiece) to handle out-of-vocabulary (OOV) words using a fixed vocabulary (e.g., ~30k tokens).

- Using the ## prefix convention from BERT's WordPiece, show how an unseen compound word like snowboarding or surfboarding is split if the vocabulary contains snow, surf, ##board, and ##ing, but not the whole words.

- Suppose you deploy a standard pre-trained BERT WordPiece tokenizer directly on a biomedical dataset containing DNA sequences (e.g., 5'-ACGATCGACTGGT-3') or complex chemical formulas (e.g., C_6H_{12}). What failure mode occurs in terms of subword fragmentation and sequence length?

- According to the diagnostic guidelines in the lecture, what specific metrics (e.g., number of subword approximations, single/double character splits, or unrecognized tokens) should you track to determine whether you need to retrain or fine-tune a domain-specific tokenizer?

Discussion Lecture: Tokenization



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Teaching note: Use today's examples (German compounds, Arabic ligatures, WordPiece) as concrete cases for the abstract questions ahead.

On the Definition of a "Word" and Linguistic Universality

In Western languages, we instinctively assume a "word" bounded by whitespace is the fundamental building block of meaning.

However, languages like Chinese or Japanese omit spaces entirely, German creates continuous compounding words like Lebensversicherungsgesellschaftsangestellter, and Arabic blends roots with complex prefixes and ligatures.

Does the concept of an isolated 'word' actually exist as an objective universal primitive of human language, or is it an arbitrary orthographic convention that forces computational pipelines to engineer artificial solutions like compound splitters and subword segmenters?

On Normalization: Every Equivalence Is a Claim About Meaning

Every normalization step (case folding, de-accenting, stemming, lemmatization) declares two different strings to be “the same”: Fed and fed, C.A.T. and cat, Müller’s and mullers.

That is not a technical operation but a claim about meaning, made once, in advance, on behalf of every future user and every future query.

Who is entitled to make that claim: the engineer who writes the rule, the corpus that shows the frequencies, or the person asking the question? And is there any representation of text that is not already an interpretation of it?

On Subword Tokenization: Morphological Insight or Statistical Convenience?

WordPiece solves the out-of-vocabulary dilemma by decomposing unknown tokens into frequent substrings (surfboard becomes surf, ##board). Outside everyday language it often produces pieces that no linguist would recognize as morphemes.

When WordPiece splits a word, is the model capturing authentic linguistic morphology (prefixes, roots, suffixes), or is it applying a statistical data-compression trick that happens to mimic linguistic structure?

If a compression trick reproduces morphology well enough to work, does the difference between “mimicking” and “capturing” structure still mean anything, and to whom?

Tutorial 01: Tokenization



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Questions?

